

The HyperTriangle

A B_4 Coxeter-Chamber Framework for Four-Axis Causal Dynamics, Spectral Closure, and Robust Certification

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Scope statement. The results in this paper close a mathematical and certification architecture *inside the declared Causal Theory (CT) model and its stated hypotheses*. “Closed” does not mean that nature has been shown to instantiate the CT postulates. Non-invasive four-sign readout and beyond-quantum empirical physics remain unestablished. The publication package accompanying Version 1.0.2 includes the verified Hyperqubit Paper-I parent archive and the Hyperqubit II CT-internal closure package. The latter contains the deterministic `proof_suite_v5.py` and its JSON receipt with 42/42 gates passing at seed 20260903. A concrete HyperTriangle application still requires owner-bound H/h or equivalent same-system receipts; those are application data, not missing theory.

Abstract

The HyperTriangle is a four-output signed-dominance geometry built from the pre-sign Hyperqubit amplitudes $h_i = o_i(\nabla\Phi_i)$. A state with nonzero, pairwise distinct magnitudes is labeled by its sign vector and the strict ordering of the four magnitudes. The resulting $2^4! = 384$ regions are exactly the chambers of the Coxeter reflection arrangement of type B_4 . After positive radial scale is removed, the state lies on the ℓ^1 unit shell, the boundary of the four-dimensional cross-polytope (16-cell), and the HyperTriangle complex is the barycentric subdivision $\text{sd}(\partial 16\text{-cell})$ with f -vector $(80, 464, 768, 384)$.

This paper develops the full declared certification chain: projective chamber dynamics and wall velocities; exact coarse-graining from 384 chambers to the 16 Hyperqubit sign states; finite-carrier and finite-spectral output closure; block-Hankel observable-order recovery; exact and certified B_4 wall-event analysis for real, Jordan, oscillatory, and near-hit spectral regimes; owner-bound reconstruction $O, U \mapsto H = OU \mapsto h = \text{diag } H$; static and transported bridge tomography; connection-derived parallel transport and holonomy constraints; perturbation propagation from numerical receipts to bridge, chamber, and route certificates; E-optimal calibration designs; and adaptive terminal stopping.

The final result is a scoped closure theorem for the HyperTriangle certification architecture. The mathematics is terminal in the declared scope, while same-system source inputs and empirical validation remain separate open inputs rather than hidden mathematical debts. The complete 52-pass reproducibility archive accompanying this paper contains the theorem ledger, executable controls, qualification campaigns, manifests, checksums, and the terminal claim firewall.

Keywords: HyperTriangle; Hyperqubit; Coxeter complex B_4 ; 16-cell; projective dynamics; spectral observability; exponential polynomials; bridge tomography; certified numerics; Causal Theory.

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1 Introduction

The HyperTriangle project began with a simple geometric question: what is the correct four-dimensional analogue of cutting a square by a diagonal into triangles? The answer is not a single four-dimensional body diagonal. The relevant structure is the complete family of pairwise coordinate-comparison hyperplanes. For four ordered quantities this produces $4! = 24$ simplex chambers per sign sector. Once the four signs are retained, the natural state space contains

$$2^4 4! = 384$$

signed-magnitude chambers.

The key physical/model input inherited from the Hyperqubit construction is not a continuous four-cube. It is the ordered family of four pre-sign amplitudes

$$h_i(t) = o_i(t)(\nabla\Phi_i(t)), \quad i = 1, \dots, 4, \quad (1)$$

with sign state

$$\Sigma(t) = (\text{sgn } h_1, \dots, \text{sgn } h_4) \in \{\pm 1\}^4.$$

The HyperTriangle refines this sign state by remembering the strict magnitude order. This refinement is mathematically useful because sign transitions and dominance swaps are different wall events, although both are generated by the same reflection arrangement.

The project subsequently expanded from geometry to a complete certification architecture. The resulting object is not merely a triangulation. It is a hierarchy connecting geometry, dynamics, observability, spectral closure, event theory, owner-bound reconstruction, frame transport, numerical uncertainty, protocol design, and a deterministic stopping rule.

The present paper is a synthesis of the terminal PASS052 archive. It is organized as a mathematical paper rather than as a chronological pass log. Internal pass identifiers are retained only where they clarify reproducibility. The supplementary archive is

HYPERTRIANGLE_AUX_CUMULATIVE_PASS052R1_20260925.zip,

with SHA-256

23767c59a02d3fe79c1bd98adb278f764a16cfc5eb434656e26e3423c09fdbea.

2 Declared model interface and claim boundary

2.1 The four-axis observable

Let

$$O(t) = \begin{pmatrix} o_1(t) \\ \vdots \\ o_4(t) \end{pmatrix}, \quad U(t) = \begin{pmatrix} \nabla\Phi_1(t) & \cdots & \nabla\Phi_4(t) \end{pmatrix}.$$

Define the full pairing matrix

$$H(t) = O(t)U(t), \quad h(t) = \text{diag } H(t). \quad (2)$$

The HyperTriangle uses h as its four continuous coordinates and $\Sigma = \text{sgn } h$ as its coarse sign projection. The off-diagonal entries of H are not decorative: in the connection-aware dynamics they can influence the future derivatives of the diagonal amplitudes.

2.2 Representation covariance

Under a simultaneous change of ambient representation

$$U \mapsto RU, \quad O \mapsto OR^{-1},$$

one has

$$H \mapsto OR^{-1}RU = H.$$

Thus H and h are representation invariant under the same co-transport rule. This observation is central later: numerical frame matrices and numerical connections may be gauge dependent even when the pairings that define the HyperTriangle are not.

2.3 What is and is not claimed

The mathematical statements below have three logically distinct statuses.

1. **Closed mathematics in declared scopes.** Exact combinatorics, linear algebra, spectral and event theorems, transport identities, perturbation inequalities, and deterministic certification algorithms.
2. **Application-specific same-system instantiation.** The mathematics is ready to consume owner-bound H/h , coframe/connection data, torus sizes, or numerical receipts. Version 1.0.2 supplies and verifies the Hyperqubit source/runtime packages; a concrete physical HyperTriangle instance still requires its own owner-bound measurements or equivalent receipts, which are inputs rather than missing theorems.
3. **Empirical realization.** The archive does not establish that nature implements the full CT postulates, does not establish a non-invasive full four-sign readout, and does not establish beyond-quantum empirical physics.

This separation is part of the theorem architecture. It is not an after-the-fact disclaimer.

3 From four signs to the B_4 HyperTriangle complex

3.1 The 24 order simplices

Consider four continuous magnitudes in one sign sector. For each permutation $\pi \in S_4$, the strict inequalities

$$|h_{\pi_1}| > |h_{\pi_2}| > |h_{\pi_3}| > |h_{\pi_4}| > 0$$

define one dominance chamber. The combinatorial precursor is the Kuhn/Freudenthal triangulation of a four-cube: one pairwise hyperplane $x_i = x_j$ exactly bisects the cube, while the six pairwise comparison hyperplanes jointly generate $4! = 24$ monotone simplices rather than 2^6 independent regions.

Each such simplex corresponds to one shortest monotone antipodal path on the four-cube sign skeleton: the permutation records the order of the four bit flips. This gives an exact bridge between the order-simplex combinatorics and the Q_4 sign skeleton without claiming that the cube interior itself is a physical Hyperqubit state space.

3.2 The reflection arrangement

The actual HyperTriangle walls in amplitude space are

$$h_i = 0, \quad |h_i| = |h_j|.$$

Equivalently,

$$h_i = 0, \quad h_i - h_j = 0, \quad h_i + h_j = 0. \quad (3)$$

There are $4 + 2\binom{4}{2} = 16$ reflection hyperplanes. These are exactly the type- B_4 reflection arrangement. Its Weyl group is the signed-permutation group

$$W(B_4) = C_2^4 \rtimes S_4, \quad |W(B_4)| = 2^4 4! = 384.$$

Theorem 3.1 (HyperTriangle chamber identification). *Away from all walls in (3), the pair consisting of the sign vector $\Sigma \in \{\pm 1\}^4$ and a strict magnitude ordering $\pi \in S_4$ labels exactly one B_4 Coxeter chamber. Hence the HyperTriangle state complex has exactly 384 open chambers.*

3.3 Projectivization and the 16-cell

Let

$$M = \|h\|_1 = \sum_i |h_i|, \quad z = \frac{h}{M}, \quad M > 0.$$

Then $\|z\|_1 = 1$, so z lies on the boundary of the four-dimensional cross-polytope, the 16-cell. Its 16 tetrahedral facets are precisely the sign sectors.

Inside each sign tetrahedron, the six equal-magnitude walls subdivide the facet into 24 tetrahedra. Therefore

$$\boxed{\mathcal{H} = \text{sd}(\partial \text{ 16-cell}) = \text{the Coxeter complex of type } B_4.} \quad (4)$$

The barycentric-subdivision f -vector is

$$\boxed{(80, 464, 768, 384)},$$

and its Euler characteristic is zero, as required for a triangulated three-sphere.

3.4 Simple-root and barycentric coordinates

Fix one chamber and list the signed magnitudes in ascending order,

$$0 < y_0 < y_1 < y_2 < y_3.$$

Define

$$\alpha_0 = y_0, \quad \alpha_1 = y_1 - y_0, \quad \alpha_2 = y_2 - y_1, \quad \alpha_3 = y_3 - y_2.$$

Then the chamber interior is $\alpha_k > 0$. Since

$$M = 4\alpha_0 + 3\alpha_1 + 2\alpha_2 + \alpha_3,$$

define

$$\boxed{\beta = \frac{(4\alpha_0, 3\alpha_1, 2\alpha_2, \alpha_3)}{M} \in \Delta^3.} \quad (5)$$

Every HyperTriangle is therefore an ordinary tetrahedral chart in projective state space. A simple wall is just $\beta_k = 0$.

4 Projective dynamics and chamber transitions

4.1 Exact radial gauge elimination

Fix a sign chamber with sign vector σ . Since $M = \sigma^T h$ on that chamber,

$$\dot{z} = \frac{1}{M}(I - z\sigma^T)\dot{h}, \quad \sigma^T \dot{z} = 0. \quad (6)$$

Consequently

$$\dot{h} = M\dot{z} + \frac{\dot{M}}{M}h.$$

Any positive rescaling $\tilde{h} = c(t)h$ leaves both z and $\dot{z}$ unchanged. Positive radial scale is therefore an exact gauge of chamber dynamics.

For the barycentric coordinates in (5), put

$$s_k = \frac{(4-k)\dot{\alpha}_k}{M}.$$

Then

$$\dot{\beta}_k = s_k - \beta_k \sum_j s_j. \quad (7)$$

At a simple wall $\beta_k = 0$, the sign of $\dot{\beta}_k$ is exactly the sign of $\dot{\alpha}_k$, so projectivization preserves the local crossing/tangency classification.

4.2 Connection-aware pairing dynamics

From $H = OU$, if

$$\nabla_\tau O = AO, \quad \nabla_\tau U = W,$$

then

$$\dot{H} = AH + OW, \quad \dot{h}_i = (AH)_{ii} + (OW)_{ii}. \quad (8)$$

In the declared local Golden sector with

$$q_i = -\Delta\Phi_i + V'_i(\Phi_i), \quad \partial_\tau \Phi_i = -K_i(\Phi_i)q_i,$$

define

$$L_{ji} = o_j(\nabla\Delta\Phi_i), \quad \lambda_i = K_i V''_i + K'_i q_i.$$

Then

$$\dot{H} = AH + LD_K - HD_\lambda. \quad (9)$$

On a sign wall $h_i = 0$,

$$\dot{h}_i = (AH)_{ii} + K_i L_{ii}. \quad (10)$$

This formula shows why the four diagonal amplitudes do not generally close the frame-transport dynamics: off-diagonal H_{ji} can affect the next wall velocity.

4.3 Exact four-event control route

A source-form heat-sector control used throughout the archive is

$$h(t) = (4e^{-t}, 3e^{-t}, 2e^{-t}, e^{-t} - 8e^{-4t}).$$

With $u = e^{-3t}$,

$$h(t) = e^{-t}(4, 3, 2, 1 - 8u),$$

so the positive radial factor is irrelevant to the B_4 route. The exact events are

u	t	simple wall	projective wall velocity
5/8	$\frac{1}{3} \log(8/5)$	s_3	-15
1/2	$\frac{1}{3} \log 2$	s_2	-12
3/8	$\frac{1}{3} \log(8/3)$	s_1	-9
1/8	$\log 2$	s_0	-3

Thus the reduced chamber word is

$$\boxed{s_3 s_2 s_1 s_0}.$$

The first three events change only magnitude order; the final event changes the Q_4 sign projection.

5 Coarse-graining, flux, and hidden dominance memory

Let $\mathcal{C}$ be the set of 384 chambers and let $r \in \Delta^{383}$ be a chamber ensemble. The deterministic projection to sign occupancy is

$$p_s = \sum_{c: \sigma(c)=s} r_c, \quad p \in \Delta^{15}.$$

Each sign has 24 chamber preimages. At a generic interior p , the hidden conditional dominance state is

$$r_{s,\pi} = p_s \kappa_{\pi|s}, \quad \kappa_{\cdot|s} \in \Delta^{23},$$

so the fixed- p fibre has dimension

$$\boxed{16 \times 23 = 368.} \tag{11}$$

Orient the 768 chamber edges and write the fine flux law

$$\dot{r} = B_{B_4} J.$$

There are 576 sign-preserving dominance edges and 192 sign-zero edges, the latter grouped into 32 Q_4 fibres of six microedges. The exact coarse factorization is

$$\boxed{PB_{B_4} = B_{Q_4} C}, \quad \boxed{\dot{p} = B_{Q_4} j, \quad j = C J.} \tag{12}$$

The kernel of instantaneous occupancy divergence has dimension

$$768 - 15 = 753 = 576 + 160 + 17.$$

Thus Δ^{15} is an exact coarse sign layer but not a complete dynamical memory state in general. The exact four-event route above projects to only one visible sign event.

6 Spectral closure and four-output observability

6.1 Finite spectral sectors

In a fixed-owner finite spectral/LTI sector, the four-output law has a finite scalar recurrence

$$q(D)h = 0 \tag{13}$$

componentwise. Consequently every B_4 wall coordinate

$$w_\alpha(t) = \alpha^T h(t), \quad \alpha \in \{e_i, e_i - e_j, e_i + e_j\},$$

satisfies the same recurrence $q(D)w_\alpha = 0$.

For a finite exponential sector

$$y_n = h^{(n)}(0) = \sum_j v_j \lambda_j^n, \quad v_j \in \mathbb{R}^4,$$

form the vector block-Hankel matrix from the jets. If the row and column delays are at least the observable rate count r_{obs} , its rank is exactly r_{obs} . This four-channel construction can reveal rates that no single scalar channel sees by itself.

A finite jet prefix alone is not terminal. For every K there exists a nonzero finite exponential sum whose first K derivatives vanish at the reference time. Therefore terminal recovery requires an order ceiling, a complete trajectory law, or an equivalent terminality owner. With a declared R_{max} and at least $2R_{\text{max}}$ exact vector jets, the terminal block-Hankel rank and the monic common recurrence are uniquely recoverable.

6.2 Finite torus ceilings

For a finite four-axis periodic carrier with

$$M = \prod_a N_a$$

sites and four scalar field components, the raw fixed-owner linear state dimension is $4M$. The four constant field directions are invisible to gradient observations; if that subspace is invariant, a safe generic ceiling is

$$\boxed{R_{\text{max}} \leq 4(M - 1).} \tag{14}$$

In the narrower common scalar-generator sector, the earlier ceiling $M - 1$ remains valid. The fact that the carrier has four axes does not by itself give a numerical ceiling; the actual N_a , M , or a smaller certified state dimension must be supplied.

6.3 Controlled spectral truncation

Let h_N be a truncated output with

$$\|h - h_N\|_1 \leq \varepsilon_h < \|h_N\|_1.$$

For projectivization $\pi(h) = h/\|h\|_1$,

$$\boxed{\|\pi(h) - \pi(h_N)\|_1 \leq \frac{2\varepsilon_h}{\|h_N\|_1 - \varepsilon_h}.} \tag{15}$$

If this certified projective error is smaller than every sign-wall margin $|z_i|$ and every pairwise magnitude gap $||z_i| - |z_j||$, then the full and truncated states lie in exactly the same HyperTriangle chamber. This creates an adaptive cutoff rule: increase the spectral cutoff until the projective error falls strictly below the current B_4 wall margin.

The archive simultaneously proves a no-go: unrestricted spectral support has no universal finite-dimensional projective closure, and nonlinear terms can generate new harmonics. The terminal architecture therefore closes exact finite sectors and certified approximation branches without asserting a universal finite closure where none exists.

7 B_4 wall-event theory

7.1 Exact algebraic isolation in commensurate sectors

Suppose

$$h(t) = \sum_j v_j e^{\lambda_j t}, \quad v_j \in \mathbb{Q}^4,$$

with rational/commensurate real rates. Let $\lambda_{\max} - \lambda_j = q_j \delta$ and set $u = e^{-\delta t}$. Then

$$h(t) = e^{\lambda_{\max} t} H(u), \quad H \in \mathbb{Q}[u]^4.$$

Every B_4 wall becomes

$$w_\alpha(t) = e^{\lambda_{\max} t} P_\alpha(u), \quad P_\alpha \in \mathbb{Q}[u].$$

Thus finite-time events reduce exactly to polynomial roots on $0 < u \leq 1$. Root multiplicity classifies transverse crossings, higher odd-order crossings, even-order tangencies, simultaneous face events, and persistent walls.

For real Jordan sectors

$$h_i(t) = \sum_j p_{ji}(t) e^{\lambda_j t}, \quad p_{ji} \in \mathbb{Q}[t],$$

the walls lie in $\mathbb{Q}[t, u]$ with $u = e^{-\delta t}$. Along the physical curve differentiation is generated by

$$D_\delta = \partial_t - \delta u \partial_u.$$

Exact factorization and certified rational enclosures yield global event isolation on the rational commensurate branch.

7.2 Arbitrary finite real rates

Commensurability is not needed for finiteness. For a nonpersistent real exponential polynomial

$$f(t) = \sum_{j=1}^r p_j(t) e^{\lambda_j t}, \quad \lambda_1 < \dots < \lambda_r \in \mathbb{R},$$

with $d_j = \deg p_j$ and

$$N = \sum_j (d_j + 1),$$

one has the global zero ceiling

$$\boxed{Z_{\mathbb{R}}(f) \leq N - 1} \tag{16}$$

with multiplicity counted. This follows by conjugated differentiation and Rolle induction.

7.3 Oscillatory spectra and infinite event routes

Genuine complex rates change the qualitative picture. After removing the dominant exponential and polynomial scale, the leading wall can reduce to a trigonometric polynomial on the closure of a linear frequency orbit. Periodic and quasiperiodic controls in the archive show that a finite-dimensional output law can then have infinitely many future B_4 crossings. Hence finite-dimensionality alone does not imply a finite chamber route.

7.4 Exponential near-hits and the critical profile

For a semidefinite dominant wall write

$$f(t) = \Theta(\omega t) + R(t), \quad \Theta \geq 0,$$

with zero set Z , local contact order κ , and lower layer

$$|R(t)| \leq C(1+t)^p e^{-\gamma t}.$$

Define the contact exponent

$$\rho_Z = \limsup_{t \rightarrow \infty} \frac{-\log d(\omega t, Z)}{t}. \quad (17)$$

Then

$$\kappa \rho_Z < \gamma$$

forces eventual dominant sign and therefore only finitely many future zeros. In the supercritical regime $\kappa \rho_Z > \gamma$, a recurrent adverse lower layer can dominate along exponentially close returns and can produce infinitely many crossings. Equality is genuinely nonterminal at the exponent level.

At the critical equality $\kappa \rho_Z = \gamma$, divide by the positive scale

$$a(t) = (1+t)^p e^{-\gamma t}$$

and write

$$f(t) = a(t)[G(t) + o(1)].$$

Then the exact terminal trichotomy is:

$$\liminf G > 0 \text{ or } \limsup G < 0 \implies \text{no sufficiently late zeros,}$$

while

$$\liminf G < 0 < \limsup G \implies \text{infinitely many zeros.}$$

Boundary-zero cases descend recursively to the next asymptotic scale or require exact-hit information. This closes the event-count architecture without forcing a verdict when the supplied asymptotic data are genuinely insufficient.

8 Owner-bound reconstruction and intrinsic-frame reduction

8.1 All-order owner jets

From (2), Leibniz gives, for every n ,

$$H^{(n)} = \sum_{a=0}^n \binom{n}{a} O^{(a)} U^{(n-a)}, \quad h^{(n)} = \text{diag } H^{(n)}. \quad (18)$$

Thus same-owner O, U at one calibrated time determine the current chamber, and owner-bound jets determine output jets exactly. The later dynamical objects A, L, K, λ are not required to *define* h ; they are needed only to validate stronger claims about why the owner evolves as asserted.

8.2 Coorientation-adapted gauge

The pairing is invariant under

$$U \mapsto RU, \quad O \mapsto OR^{-1}.$$

In a local rank-four sector, choose $R = O$. Then

$$O_{\text{adapted}} = I, \quad U_{\text{adapted}} = H.$$

Therefore H and all of its jets are the intrinsic owner-bound field coordinates in the coorientation frame. A numerical standalone coorientation connection is representation dependent: a time-dependent gauge can change its matrix while leaving all pairings fixed. The irreducible object for HyperTriangle observation is the owner-bound relative pairing/field content, not a preferred coordinate connection matrix.

9 Coframe bridge, tomography, and parallel transport

9.1 Static bridge

Let E be a source coframe on the same rank-four carrier as O . Then there is a unique

$$B \in GL(4)$$

with

$$O = BE. \tag{19}$$

For the gradient probe matrix U , define

$$J = EU, \quad H = OU.$$

Then

$$\boxed{H = BJ.} \tag{20}$$

If J is invertible,

$$\boxed{B = HJ^{-1}.} \tag{21}$$

If both frames are orthonormal for the same positive metric and share orientation, $B \in SO(4)$. Three independent full probe correspondences then determine the oriented orthogonal bridge uniquely, while two leave an $SO(2)$ ambiguity. Four signs alone never calibrate a continuous bridge.

9.2 Diagonal-amplitude tomography

Full H is not necessary during calibration. Suppose a constant bridge is probed with known matrices $J^{(s)}$ and only the four diagonal amplitudes

$$h_i^{(s)} = (BJ^{(s)})_{ii}$$

are measured. For each output i , let

$$W_i[s, k] = J_{ki}^{(s)}.$$

Then the i th row of B is the unique solution precisely when

$$\boxed{\text{rank } W_i = 4 \quad \text{for every } i.} \quad (22)$$

Four preparations are necessary in unrestricted $GL(4)$ because each preparation supplies one equation per four-unknown row. They are also sufficient. With the cyclic permutation matrix P and

$$J^{(s)} = P^s, \quad s = 0, 1, 2, 3,$$

one obtains

$$h_i^{(s)} = B_{i, i+s \pmod{4}},$$

so the four preparations read all sixteen bridge entries exactly once.

9.3 Transported bridge

Let

$$B(t_s) = L_s B_0 R_s^{-1}. \quad (23)$$

With $K_s = R_s^{-1} J_s$, the scalar measurements become

$$h_{s,i} = e_i^T L_s B_0 K_s e_i = a_{s,i}^T B_0 b_{s,i}.$$

After vectorization,

$$\boxed{D \text{ vec}(B_0) = d.} \quad (24)$$

The exact terminal criterion is

$$\boxed{B_0 \text{ is uniquely identifiable} \iff \text{rank } D = 16.} \quad (25)$$

The static calibration is the special case $L_s = R_s = I$.

9.4 Connection-derived transport and holonomy

For either frame owner $X \in \{O, E\}$, define the fundamental transport along a declared path by

$$\dot{T}_X(t, t_0) = -\Omega_X(t) T_X(t, t_0), \quad T_X(t_0, t_0) = I. \quad (26)$$

Formally,

$$T_X(t, t_0) = \mathcal{P} \exp \left[- \int_{t_0}^t \Omega_X(\tau) d\tau \right].$$

With $L = T_O$ and $R = T_E$, equation (23) follows from the bridge-intertwining law. The transport satisfies the cocycle and inverse identities. Time ordering is irreducible: two noncommuting segments can have the same integrated matrix and different transports.

For a loop based at the reference point, a connection-preserving bridge must satisfy the holonomy condition

$$\boxed{W_O B_0 = B_0 W_E.} \quad (27)$$

Hence path data cannot be discarded when curvature is nonzero.

10 Robust certification from numerical receipts

10.1 Rank stability under transport uncertainty

Suppose

$$\|L - \hat{L}\|_2 \leq \varepsilon_L, \quad \|R - \hat{R}\|_2 \leq \varepsilon_R.$$

If

$$\|\hat{R}^{-1}\|_{2\varepsilon_R} < 1,$$

then

$$\|R^{-1} - \hat{R}^{-1}\|_2 \leq \frac{\|\hat{R}^{-1}\|_{2\varepsilon_R}^2}{1 - \|\hat{R}^{-1}\|_{2\varepsilon_R}}. \quad (28)$$

Each design row has the form $d = a \otimes b$. If $\|\delta a\| \leq \varepsilon_a$ and $\|\delta b\| \leq \varepsilon_b$, then

$$\|d - \hat{d}\|_2 \leq \varepsilon_a \|\hat{b}\|_2 + \|\hat{a}\|_2 \varepsilon_b + \varepsilon_a \varepsilon_b. \quad (29)$$

For row bounds δ_r , define

$$\eta_D = \left(\sum_r \delta_r^2 \right)^{1/2}.$$

Then

$$\eta_D < \sigma_{\min}(\hat{D}) \implies \text{rank } D = 16 \quad (30)$$

for every true design inside the declared uncertainty balls.

Once a bridge-error ball is obtained, it propagates to $H = BJ$ and to componentwise intervals

$$h_i \in [\hat{h}_i - \varepsilon_i, \hat{h}_i + \varepsilon_i].$$

A chamber is certified unchanged when

$$|\hat{h}_i| > \varepsilon_i \quad (31)$$

and

$$||\hat{h}_i| - |\hat{h}_j|| > \varepsilon_i + \varepsilon_j \quad (32)$$

for all relevant indices.

10.2 From solver receipts to error budgets

If a nominal connection $\hat{\Omega}$ satisfies

$$\|\hat{\Omega}\| \leq \hat{M}, \quad \|\Omega - \hat{\Omega}\| \leq \varepsilon_\Omega$$

on a path interval of length Δ , then

$$\|T_\Omega - T_{\hat{\Omega}}\| \leq \varepsilon_\Omega \Delta \exp[(\hat{M} + \varepsilon_\Omega)\Delta]. \quad (33)$$

An order- N Peano–Baker truncation has the safe tail

$$\varepsilon_{PB} \leq e^{\hat{M}\Delta} \frac{(\hat{M}\Delta)^{N+1}}{(N+1)!}. \quad (34)$$

If a numerical transport solver supplies residual

$$r = \dot{\hat{T}} + \hat{\Omega}\hat{T},$$

with $\|\Omega\| \leq M$, $\|\Omega - \hat{\Omega}\| \leq \varepsilon_\Omega$, $\|\hat{T}\| \leq Q$, $\|r\| \leq \varepsilon_r$, and initial error ε_0 , then Gronwall gives

$$\boxed{\|T - \hat{T}\| \leq e^{M\Delta}[\varepsilon_0 + \Delta(\varepsilon_\Omega Q + \varepsilon_r)]}. \quad (35)$$

For the observation $h_i = o_i(u_i)$, separate coorientation, gradient, and readout errors give

$$\boxed{\varepsilon_{h_i} \leq \varepsilon_{o_i}\|\hat{u}_i\| + \|\hat{o}_i\|\varepsilon_{u_i} + \varepsilon_{o_i}\varepsilon_{u_i} + \varepsilon_{\text{read}}}. \quad (36)$$

Analogous bounds on $\dot{h}$ produce wall-velocity budgets. Thus PASS049 closes a receipt-to-certificate interface without inventing historical numerical tolerances that are absent from the source corpus.

11 Optimal calibration design and adaptive terminal stopping

11.1 E-optimality

For an $m \times 16$ design matrix D , let

$$E = \|D\|_F^2.$$

The mean eigenvalue of $D^T D$ is $E/16$, hence

$$\boxed{\sigma_{\min}(D)^2 \leq \frac{E}{16}, \quad \sigma_{\min}(D) \leq \sqrt{E/16}}. \quad (37)$$

Equality holds exactly when

$$\boxed{D^T D = \frac{E}{16} I_{16}}. \quad (38)$$

Thus the E-optimal protocols are tight frames in the 16-dimensional bridge-parameter space.

The four-preparation cyclic design is exactly such a tight frame. Its sixteen rows are a permutation of the standard basis of $\mathbb{R}^{16}$, so

$$D^T D = I_{16}, \quad \sigma_{\min} = \sigma_{\max} = 1, \quad \kappa_2(D) = 1.$$

Under relative row uncertainty $\delta_r \leq \varepsilon\|d_r\|$, a tight frame obeys

$$\sigma_{\min}(D) - \eta_D \geq \|D\|_F \left(\frac{1}{4} - \varepsilon \right). \quad (39)$$

The worst-case Frobenius certificate therefore has the sharp positivity threshold $\varepsilon < 1/4$.

If right transport is compensated by probes $J_s = RP^s$, then

$$D^T D = (L^T L) \otimes I_4.$$

In the metric branch where L is orthogonal, perfect conditioning survives nontrivial right transport exactly.

11.2 Redundancy and dropout

Repeating the optimal 16-row block K times gives

$$D^T D = K I_{16}, \quad \sigma_{\min} = \sqrt{K}.$$

After arbitrary loss of $q < K$ rows,

$$\boxed{\sigma_{\min} \geq \sqrt{K - q}.} \quad (40)$$

For $q \geq K$, no universal rank guarantee remains because all copies of one parameter direction may be lost.

11.3 First-terminal-prefix stopping

Let an ordered stream of certified measurement receipts be $S_1, S_2, \dots$. For the prefix through n , define the terminal predicate P_n to require simultaneously:

1. robust unique bridge calibration;
2. stable B_4 chamber certificates for every declared target;
3. certified unique wall crossings for every declared event;
4. strictly ordered, disjoint certified event-time intervals.

If the set of terminal prefixes is nonempty, define

$$\boxed{n_* = \min\{n : P_n = \text{true}\}.} \quad (41)$$

In the unrestricted $GL(4)$ diagonal-amplitude interface one receipt contributes at most four scalar equations, so $\text{rank } D \leq 4n$ and $n \geq 4$ is necessary. The cyclic construction attains $n = 4$, proving

$$\boxed{n_{\text{calibration}, \min} = 4}$$

for calibration alone.

Terminal B_4 certification may require more receipts because the target can be arbitrarily close to a wall or the uncertainty budgets can be too large. Moreover, terminality is not monotone under arbitrary receipt appending: a low-information, high-uncertainty receipt can increase the aggregate worst-case error enough to destroy a previously positive robust margin. The deterministic policy is therefore to stop at the first terminal prefix or to screen a candidate receipt before accepting it.

12 Terminal closure theorem

Theorem 12.1 (Scoped closure of the HyperTriangle architecture). *Within the declared hypotheses and claim firewalls of the cumulative archive, no unresolved internal mathematical dependency remains that is required to execute the HyperTriangle certification chain. The following blocks are closed in their declared scopes: B_4 geometry, projective dynamics, coarse-graining and memory accounting, finite-sector spectral/observable closure and certified truncation, wall-event theory, owner-to-observable reconstruction, static and transported bridge tomography, connection-derived transport, robust uncertainty propagation, receipt-to-budget conversion, E -optimal design, and adaptive terminal stopping.*

The theorem is intentionally scoped. It does not assert a universal finite-dimensional closure for arbitrary nonlinear Golden Flow. The archive contains explicit no-go theorems that reject such an overstatement. Nor does closure of the certification architecture supply absent physical inputs automatically.

12.1 Application-specific source/same-system inputs

The terminal audit retains the following as inputs to already-closed mathematics rather than as missing theorems:

- for a concrete physical instance, owner-bound H/h measurements or an equivalent same-system receipt stream;
- direct intrinsic H/h data or the same-system field/connection/observation data that generate them;
- actual finite-discrete torus sizes when a numerical carrier ceiling is needed;
- modal-tail/Galerkin constants for the low-cutoff approximation branch;
- a physical common-scale contract for the four continuous pairings when the corresponding dominance interpretation is invoked;
- same-system contraction and wall-regularity constants for the approximation branch.

Exact owner-bound H/h can bypass much of the bridge/connection route.

12.2 Empirical boundary

The terminal archive records the following statuses verbatim in substance:

CT empirical realization = NOT ESTABLISHED,

non-invasive full four-sign readout = NOT ESTABLISHED,

beyond-QM empirical physics = NOT ESTABLISHED.

Version 1.0.2 directly includes and verifies the Hyperqubit provenance packages. The supplied Paper-I parent archive is `QBIT_HQ_FINAL_V1P004_WINDOWS.zip` with SHA-256 `2146276c3339e2cb1683874049b01066922c926b8d6099be28356be439c7ee49`. The supplied Hyperqubit II CT-internal closure package has SHA-256 `57a6238f03ae0e62bf5c6b772e9c8f451e891965380358b34ace874c628035dd`; it contains `proofs/proof_suite_v5.py` and `proofs/PROOF_SUITE_V5_RESULTS.json`. The receipt reports 42/42 PASS with seed 20260903, and the embedded Paper-I parent is byte-identical to the separately supplied parent archive. Therefore no Hyperqubit proof-suite artifact is missing from the publication provenance. Owner-bound H/h for any particular physical HyperTriangle application remains an application input, not a closure debt.

13 Reproducibility and computational qualification

The cumulative PASS052R1 archive contains one research note for every pass 001–052, the theorem/reduction/no-go ledger, typed schemas, exact controls, independent verifiers, qualification campaigns, source-binding audits, claim firewalls, deterministic manifests/checksums, and a provenance-recovery addendum. Historical pass notes are preserved as records of what was visible during each pass; the PASS052R1 addendum supersedes only the terminal provenance

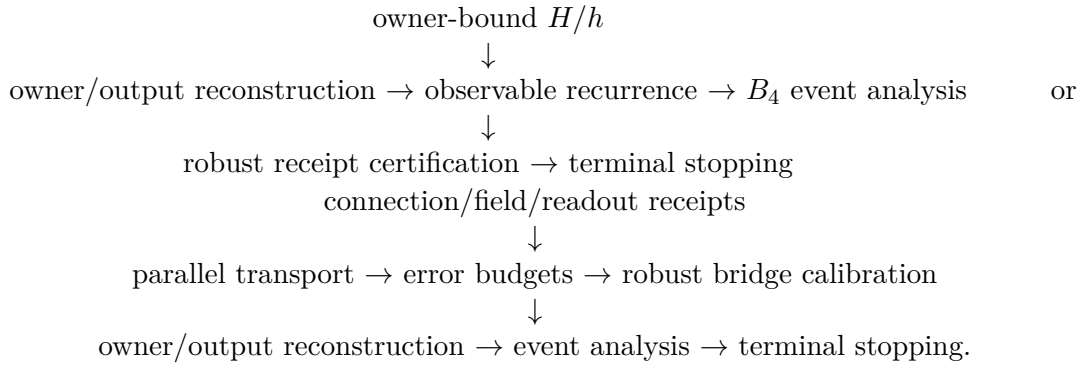
status. The Zenodo publication package also ships the two verified Hyperqubit source packages and exposes the proof suite and 42-gate JSON receipt for direct inspection.

The terminal validation records:

Terminal unit audit	59/59 PASS
Independent campaign	250001/250001 PASS
Qualification campaign	600000/600000 PASS
PASS051 regression	86/86 PASS
Parent nonmutable integrity	1150/1150 PASS
Cold replay	PASS
Base PASS052 final manifest entries	1170
Base PASS052 final checksum entries	1169
PASS052R1 provenance validation	7/7 PASS
PASS052R1 manifest entries	1174
PASS052R1 checksum entries	1173

These computational campaigns are implementation and consistency witnesses. They are not substitutes for external empirical evidence.

The executable operational routes are, schematically,



14 Discussion

The main conceptual consequence of the HyperTriangle construction is that the four Hyperqubit signs are only the coarsest discrete shadow of a richer signed-dominance geometry. Projectivization removes an irrelevant positive radial degree of freedom and reveals a canonical finite complex: the B_4 Coxeter complex on the boundary of the 16-cell. The 384 chambers are therefore not an arbitrary discretization chosen for numerical convenience; they are fixed by the four sign hyperplanes together with the twelve equal-magnitude reflection hyperplanes.

The second consequence is methodological. Geometry alone does not close dynamics. The project therefore tracks exactly what information is lost under coarse-graining, what memory is needed for future wall motion, when finite spectral recurrence is available, and when it is impossible. The same discipline is maintained for physical/source binding: a convenient coordinate frame is not silently promoted to an intrinsic physical frame, and application-specific data are not confused with missing theory.

The third consequence is operational. Once owner-bound data or certified numerical receipts are supplied, the later stages are no longer informal: rank, recurrence, event multiplicity, bridge identifiability, transport uncertainty, chamber margins, route ordering, protocol conditioning,

and stopping all have explicit decision criteria. This is the sense in which the HyperTriangle architecture is closed.

15 Conclusion

The HyperTriangle turns four continuous signed pairings into a canonical 384-chamber projective geometry and then builds a certification chain around that geometry. The final architecture can distinguish sign and dominance events, preserve the hidden information discarded by the 16-sign projection, exploit finite spectral or carrier structure when available, reject universal finite closure when it is not, reconstruct the observable from owner-bound data, calibrate frame bridges from diagonal amplitudes, derive transport from connections, propagate numerical uncertainty to chamber and event certificates, choose E-optimal measurements, and stop at the first terminal receipt prefix.

Accordingly, the terminal state of the project is

HYPERTRIANGLE MATHEMATICAL/CERTIFICATION ARCHITECTURE = CLOSED

within the declared scope, together with

CONCRETE SAME-SYSTEM INSTANTIATION = APPLICATION INPUT

and

EMPIRICAL VALIDATION = NOT ESTABLISHED.

No automatic successor pass is required. New HyperTriangle work is justified only by a contradiction, a deliberate scope extension, new application-specific owner-bound or empirical data, or an explicit external integration requirement.

A Terminal frontier classification

The terminal audit separates residual items into two nonblocking classes.

Source or same-system application inputs: owner-bound H/h or equivalent receipts for a concrete system; intrinsic field owner or connection-derived transport data when that route is used; numerical discrete torus size where needed; low-cutoff spectral/Galerkin certificate data; the physical common-scale binding for hyperdiagonal dominance; and same-system contraction/boundary-regularity constants for the approximation branch.

Deferred nonblocking items: CT taxonomy binding, inherited active-metric/source-route materialization, and future AC-core integration. These may matter to neighboring projects but do not block the HyperTriangle certification architecture.

B Source corpus and provenance

The mathematical synthesis in this paper is grounded in the cumulative HyperTriangle archive and its bounded source corpus. The principal inherited model documents are:

1. S. D. Bolduc, *Hyperqubit II / Golden Flow*. Source file in the cumulative archive: 1054HYPERQUBIT_II_GOLDEN_FLOW_EN.pdf.
2. S. D. Bolduc, *Causal Time*, source file 14Causal Time.pdf.

3. S. D. Bolduc, *Causal Action*, source file `896causal action.pdf`.
4. S. D. Bolduc, *HyperTriangle AUX cumulative research archive*, *PASS001–PASS052R1*, 2026. Provenance-revised supplementary archive SHA-256: `23767c59a02d3fe79c1bd98adb278f764a16cfc5eb434656e26e3423c09fdbea`.
5. S. D. Bolduc, *Qubit/Hyperqubit Program Paper-I parent archive*, *QBIT_HQ_FINAL_V1P004_WINDOWS.zip*, SHA-256 `2146276c3339e2cb1683874049b01066922c926b8d6099be28356be439c7ee49`.
6. S. D. Bolduc, *Hyperqubit II Golden Flow CT-internal closure proof package*, SHA-256 `57a6238f03ae0e62bf5c6b772e9c8f451e891965380358b34ace874c628035dd`; contains `proofs/proof_suite_v5.py` and the 42-gate JSON receipt.

Classical mathematical tools used inside the derivations include Coxeter reflection geometry, Cayley–Hamilton theory, Vandermonde/block-Hankel factorization, Rolle induction, Hermite–Lindemann transcendence, Kronecker torus recurrence, Peano–Baker series, Duhamel variation of constants, Weyl singular-value perturbation, and Gronwall inequalities. Their use does not constitute an additional CT physical postulate.